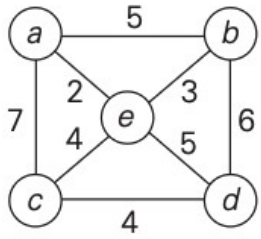
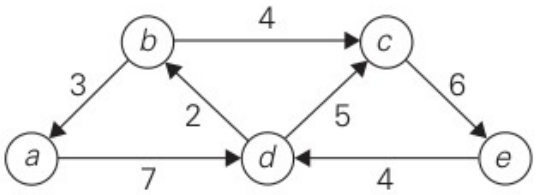
Analysis of Algorithms
Lab 09 – Greedy Technique

1. Apply Prim's algorithm to the following graph. Include in the priority queue all the vertices not already in the tree.

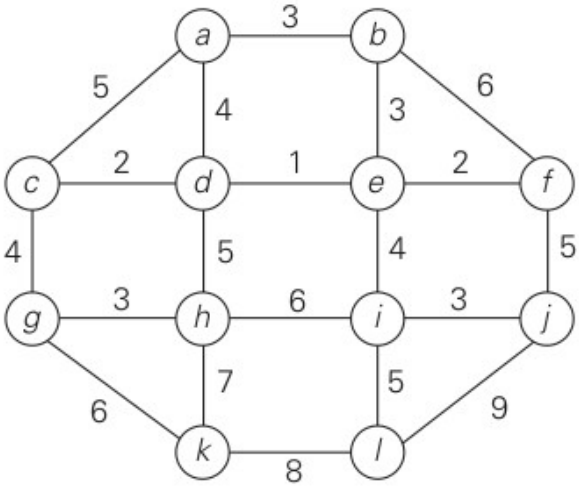


2. Create a C++ program that implementes the Prim's Algorithms. Show an execution example.
3. Solve (paper and pencil) the following instances of the single-source shortest-paths problem with vertex *a* as the source using Dijkstra's Algorithm:

   a.



   b.



4. Create a C++ program that implementes the Dijkstra's Algorithms. Show and execution example.
5. Huffman Code.
   5.1. Construct a Huffman code for the following data:

symbol	A	B	C	D	_
frequency	0.4	0.1	0.2	0.15	0.15

   5.2. Encode ABACABAD using the code of the question
   5.3. Decode 100010111001010 using the code of the question

Submission

Upload your solutions as a PDF file. In case of coding assignments, upload to GitHub and include the link in the PDF file. Your submission should include screenshots of the execution of your code when applicable.